A Lean Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

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Abstract

We report on a first Lean formalization of the logical corridor from IUT III, Theorem 3.11 to Corollary 3.12. The formalization is deliberately neutral with respect to the correctness of IUT as a proof of the abc conjecture. It isolates the ordered-real comparison required at the end of Corollary 3.12, tracks the finite q^{j^2} and label-quotient phenomena that enter the public dispute, and records exactly which source-facing construction hypotheses are still external.

The current code proves conditional theorems. If the hull/log-volume upper-ray comparison supplies $-|\log(q)| \leq -|\log(\Theta)|$, and if the displayed hypothesis $-|\log(\Theta)| \leq C_\Theta|\log(q)|$ is supplied with $|\log(q)| > 0$, then Lean proves $C_\Theta \geq -1$. If the upper-ray membership is strict, Lean proves $C_\Theta > -1$. In the boundary case $C_\Theta = -1$, Lean proves equality of the q-pilot reading and the Θ-hull boundary, and then classifies the local Frobenioid p_v^N-shift behavior under the actual upper-ray quotient. This does not prove Corollary 3.12 from Mochizuki's published hypotheses; it identifies the remaining mathematical construction problem.

After a source audit against IUT I–IV, Mochizuki's 2026 LeanForm report, and Scholze–Stix, the status is unchanged in kind but sharper in formulation: the final ordered-real implication is formalized, and several collapse-prone comparison levels are separated, but the code still assumes the source construction of the multiradial/HDD/SHE/IPL data that IUT III assigns to Theorem 3.11 and its reorganization.

1 Aim and Sources

The project formalizes a narrow and contested part of Mochizuki's inter-universal Teichmuller theory. The source base is the local copy of IUT I–IV, Mochizuki's 2026 report on formalization, and the Scholze–Stix note. The current target is the portion that Mochizuki's report calls Stage 1:

$$\text{IUT III, Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{IUT III, Corollary 3.12.}$$

The notation “3.11.5” is not a statement in IUT III. It is Mochizuki's 2026 name for the reorganized boundary where the hull-plus-determinant operation is moved before the final simultaneous comparison of the Θ-pilot and the q-pilot.

The formalization is source-facing but not source-complete. It does not construct Hodge theaters, Frobenioids, log-shells, prime strips, or theta links from first principles. Instead, these are represented by typed records with named hypotheses. The point of the current work is to test whether the claimed formal corridor is coherent once every comparison level, label quotient, ordered real line, hull boundary, and quotient collapse is made explicit.

2 Framework and Tooling

The repository is organized as a Lean verification project, not as a prose annotation project. The source papers in `docs/` are treated as the mathematical source of truth. The formalization then moves through three layers:

$$\begin{gathered}\text{source papers}\\ \Downarrow\\ \text{Lean definitions, records, and theorems}\\ \Downarrow\\ \text{this paper draft.}\end{gathered}$$

The paper draft records only the current mathematical state of the formalization. It is not a research log.

The Lean code uses typed interfaces to prevent comparison-level drift. A datum that is merely a representative j^2-coordinate computation has a different type from a balanced full-label datum, and both differ from the final hull/log-volume comparison datum. Likewise, ordered real-line transports, Hodge/SHE preservation records, upper-ray membership, and quotient-collapse audits are separate structures. This is the main anti-collapse mechanism in the framework: a proof cannot silently use a raw coordinate equality where the Corollary 3.12 corridor asks for a hull/log-volume comparison.

The module roles are:

- `IUTStage1SourceCore`: core real-line, pilot, hull, quotient, and C_Θ algebra.
- `IUTStage1Remark312Absorption`: Remark 3.12.2(v) absorption interfaces.
- `IUTStage1Source`: finite labels, Gaussian data, SHE/Hodge interfaces, and route theorems.
- `IUTStage1Experiments`: compiled endpoint checks and diagnostic theorems.

The experiment file is not a substitute for proof. It is a curated surface that forces the important route theorems to elaborate together and exposes Boolean/report endpoints such as whether the present stage settles the dispute. At present that endpoint is deliberately false.

The build discipline is simple. Each mathematical increment is checked by Lean via `lake build Iut.Stage1.IUTStage1Experiments`; the paper is rebuilt from this LaTeX source; and the code is scanned for explicit proof placeholders such as `sorry`, `admit`, and `axiom`. The long-term collaboration plan is compatible with Apodixis: the current typed route endpoints and missing-obligation records give a natural surface for a collaboration tool to show which mathematical interfaces are proved and which remain open.

3 Dispute Interface

In IUT III, Step (x) treats the procession-normalized log-volume as unchanged under (Ind1) and (Ind2), while (Ind3) supplies a one-sided upper-semi comparison. Step (xi) then passes through multiradiality, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the q-pilot reading in the upper ray determined by the Θ-pilot:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}.$$

The final numerical step is elementary:

$$-|\log(q)| \leq -|\log(\Theta)| \leq C_\Theta |\log(q)|, \qquad |\log(q)| > 0 \quad \Longrightarrow \quad C_\Theta \geq -1.$$

Scholze–Stix object to the comparison because the Θ-side data carries j^2-scaled degrees, while a naive identification of ordered real lines appears to remove the room needed for these scalars. The formalization therefore separates four comparison levels:

representative j^2 data
balanced sign-compatible data
averaged cusp-label data
hull/log-volume data.

The final route consumes only the hull/log-volume comparison level. Lean rejects the balanced sign-compatible level as sufficient for the final comparison.

4 Source Audit

The local sources impose the following constraints on the formalization. IUT I constructs the initial Θ-data and the $\Theta^{\pm \mathrm{ell}}$NF-Hodge theaters. It is not merely a notation layer: the paper's central point is that the Hodge theaters separate the additive and multiplicative combinatorial dimensions of a number field. IUT II supplies the Hodge–Arakelov theta evaluation, theta values at ℓ-torsion points, Gaussian Frobenioids, conjugate synchronization, and the multiradial/coric phenomena needed later. The current code models only finite label and Gaussian degree shadows of this material.

IUT III is the direct source for the formal corridor. Step (x) says that procession-normalized log-volumes are invariant under (Ind1) and (Ind2), while (Ind3) becomes an upper-semi inequality after applying log-volume. Step (xi) uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and the 1-column log-Kummer correspondence to put the q-pilot log-volume in the upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$. Step (xii) warns that replacing global realified Frobenioids by local Frobenioids loses calibration because local objects admit p_v^N-type endomorphisms. Remark 3.12.2 further stresses that the argument depends on weakened data, distinct q/Θ labels, upper-semi indeterminacy, and the asymmetry between the 0-column Θ-pilot and the 1-column q-pilot.

IUT IV uses the resulting log-volume estimate as input to elementary log-volume and Diophantine inequalities. Our IUT IV code is therefore only conditional algebra at present: it records how later estimates consume a Corollary 3.12-shaped bound, but it does not prove that bound from the full arithmetic constructions.

Mochizuki's 2026 report explicitly recommends Stage 1 as the implication *IUT* III, Theorem 3.11 $\Rightarrow$ Corollary 3.12, with a reorganized intermediate "3.11.5" obtained by moving the hull-plus-determinant operation before the final simultaneous comparison. The same report states that the current official code is skeletal. Our formalization is best read as a more diagnostic Stage 1 interface: it does not replace Stage 2, because it has not constructed Theorem 3.11.

Scholze–Stix argue that, after simplification, the comparison loses the j^2-scaled theta contribution and becomes essentially contentless. The Lean code therefore treats the j^2-representative level, sign quotient, averages, square/full-label preservation, ordered real-line transports, and hull/log-volume endpoint as different types. This confirms that the final Corollary 3.12 endpoint cannot be obtained from the balanced sign-compatible level alone. It does not prove that Mochizuki's source constructions supply the missing higher-level data.

5 Formalized Corridor

The relevant implementation lives in four Lean modules:

- `Iut/Stage1/IUTStage1SourceCore.lean`
- `Iut/Stage1/IUTStage1Remark312Absorption.lean`
- `Iut/Stage1/IUTStage1Source.lean`
- `Iut/Stage1/IUTStage1Experiments.lean`

5.1 Finite label model

The code defines a finite $\mathbb{F}_\ell$-label model, its sign quotient $|F_\ell| = F_\ell/\{\pm 1\}$, and the canonical half-range representatives

$$j = 0, \ldots, (\ell - 1)/2.$$

It proves:

$$\deg(\Theta_j) = j^2 \deg(q)$$

for the degree-level Gaussian endpoint, with sign invariance on the full-label quotient. It also proves that $j = 1$ and $j = 2$ give distinct degrees exactly when $\deg(q) \neq 0$. Thus the q^{j^2} phenomenon is present in the model; it is not hidden inside a scalar identification.

The coordinate average, the nonzero-coordinate average, and the absolute-label average are all formalized separately. Lean proves the exact denominator relations and shows that these averages differ strictly for nonzero environment degree. This prevents silently replacing the paper's ℓ-denominator convention by an auxiliary nonzero-only average.

5.2 Indeterminacy and upper-semi route

The Step (x) corridor is represented as:

$$A_0 \quad = \quad A_1 \quad = \quad A_2 \quad \leq U_{\mathrm{Ind3}},$$

where A_0, A_1, A_2 are the before-, after-(Ind1)-, and after-(Ind2)-averaged log-volumes, and U_{Ind3} is the (Ind3) upper-semi target. From this data and the hull/determinant identifications, the code constructs the Step (xi) upper-ray datum:

$$q_{\mathrm{avg}} \leq \Theta_{\mathrm{hull}}.$$

The same route constructs the two q-pilot readings in Step (xi-g) and proves that they are equal to the same fixed real value.

5.3 SHE and Hodge-descent boundary

The current SHE layer is conditional. It has typed records for:

structured SHE square-weight transport
factored square/full-label preservation
Hodge-descent packet transport
nonarchimedean (Ind3) source-target alignment.

Given these inputs, Lean constructs the weighted theta comparison consumed by the final route:

$$q_{\mathrm{signed}} \leq \Theta_{\mathrm{signed}}.$$

The nonarchimedean (Ind3) entry, factored square/full-label preservation, and Hodge-descent packet transport now compose directly to this comparison. With the q-pilot positivity condition and the displayed C_Θ-bound, the same composed route instantiates the signed C_Θ endpoint and proves $C_\Theta \geq -1$. The instantiated endpoint also exposes the strict and boundary alternatives: strict source comparison gives $C_\Theta > -1$, while $C_\Theta = -1$ forces equality of the signed q- and Θ-readings and makes the Θ-reading negative. These facts are now also packaged as a single route-level alternative: boundary equality with negative Θ-reading, or strict $C_\Theta > -1$. The code also proves a Gaussian-to-factored-SHE bridge at the present finite degree level. In particular, equality of the source and target environment degrees gives full-label value preservation; nonzero-label target bounds are enough because the zero label has square weight zero. The newest route uses this bridge before applying the nonarchimedean endpoint: Gaussian degree evaluations with preserved environment degree now construct the factored square/full-label preservation package consumed by the nonarchimedean C_Θ dichotomy. This removes one primitive preservation input from that route, but it still assumes the nonarchimedean (Ind3) alignment and the Hodge-descent/SHE source audit. There is also an identity-coordinate specialization. In that case equality of Gaussian degree at the canonical full label 1 implies preservation of the environment degree, so the nonarchimedean C_Θ dichotomy no longer takes environment-degree preservation as an independent hypothesis. The raw nonarchimedean entry alignment has also been refactored through a source-facing 1-column log-Kummer upper-semi compatibility object. This records the q-pilot column and log-Kummer non-interference before converting to the raw real equalities consumed by the current route.

5.4 Step (xi-f) algebra

The final real algebra is fully formalized. In informal notation, the main theorem is:

Theorem 1 (Formal Step (xi-f))**.** *Assume $q < 0$, $q \leq \theta$, and $\theta \leq C(-q)$. Then $C \geq -1$. If $q < \theta$, then $C > -1$. If $C = -1$, then $q = \theta$ and $\theta < 0$.*

This theorem is instantiated by the Step (x)-to-Step (xi) route whenever the hull/log-volume upper-ray comparison and the displayed C_Θ-bound are available. It is also instantiated by the current nonarchimedean (Ind3)-plus-factored-SHE route, subject to the same positivity and displayed-bound hypotheses. Thus the remaining gap is not the final ordered-real implication, including its strict and boundary branches, but the construction of the source hypotheses consumed by that implication.

6 Boundary Quotient Experiment

The most developed diagnostic concerns Step (xii), where local Frobenioid p_v^N-ambiguity is compared with the globally calibrated log-volume. The formal model records a local shift

$$L_N = L_0 + N\delta, \qquad N \in \mathbb{Z},\ \delta \in \mathbb{R},$$

and a global calibration that fixes exponent 0. Lean proves

$$L_N = L_0 \iff N\delta = 0,$$

and, if $\delta \neq 0$,

$$L_N = L_M \iff N = M.$$

At the boundary branch $C_\Theta = -1$, the formal corridor identifies

$$q_{\text{pilot}} = \Theta_{\text{hull}} = L_0.$$

For the actual upper-ray quotient that collapses $\{x \mid x \le \Theta_{\text{hull}}\}$, Lean proves the exact classification

$$[L_N] = [q_{\text{pilot}}] \quad \Longleftrightarrow \quad N\delta \le 0,$$

and the complementary separation criterion

$$[L_N] \ne [q_{\text{pilot}}] \quad \Longleftrightarrow \quad N\delta > 0.$$

If $\delta > 0$, this becomes:

$$[L_N] = [q_{\text{pilot}}] \Longleftrightarrow N \le 0, \qquad [L_N] \ne [q_{\text{pilot}}] \Longleftrightarrow 0 < N.$$

This is not a proof for either side of the Mochizuki–Scholze–Stix dispute. It is a precise reading of one simplified quotient mechanism inside the current formal corridor. It shows that quotient collapse can identify unequal underlying log-volumes if they lie in the collapsed upper ray, while positive local shifts remain separated. The decisive question is therefore not the ordered-real algebra; it is whether the source IUT constructions actually supply the required calibrated hull/log-volume comparison without making an illicit identification of histories or comparison levels.

7 Current Results

The project currently proves the following source-facing results in Lean:

1. The finite q^{j^2} degree rule, sign quotient, half-range representatives, zero/nonzero split, and average formulas are formalized.

2. A single label-independent scalar cannot absorb all representative j^2-degrees when $\deg(q) \ne 0$.

3. The raw representative branch does not descend to the sign quotient, while the balanced full-label Gaussian branch does.

4. The Step (x) averaged-log-volume corridor feeds a Step (xi) upper-ray comparison once determinant/hull identifications are supplied.

5. A nonarchimedean (Ind3) entry with factored square/full-label preservation now directly supplies the final weighted-theta route and the signed $C_\Theta \ge -1$ endpoint, once q-positivity and the displayed C_Θ-bound are supplied.

6. The same nonarchimedean route now exposes the strict branch $C_\Theta > -1$ and the boundary consequence $C_\Theta = -1 \Rightarrow q_{\text{signed}} = \Theta_{\text{signed}} < 0$. It also packages these as one dichotomy at the route level.

7. Gaussian degree evaluations with preserved environment degree now feed directly into this nonarchimedean C_Θ dichotomy, constructing the factored square/full-label SHE package internally.

8. In the identity-coordinate Gaussian route, equality at the canonical full label 1 supplies the environment-degree preservation needed by the same nonarchimedean C_Θ dichotomy.

9. A 1-column log-Kummer upper-semi compatibility object now supplies the raw nonarchimedean (Ind3) entry alignment used by the canonical Gaussian route.

10. The Step (xi-f) real algebra proves $C_\Theta \geq -1$, strictness under strict upper-ray membership, and equality constraints in the boundary case.

11. The final route rejects the balanced sign-compatible comparison level as sufficient; it requires hull/log-volume comparison data.

12. The local Frobenioid shift quotient experiment gives an exact threshold for collapse versus separation in the boundary upper ray.

13. Several downstream IUT IV numerical estimates are formalized as ordered-real algebra conditional on the Corollary 3.12 input.

The first-pass experiment report evaluates the route-theorem, finite Gaussian, collapse-obstruction, upper-ray, and IUT IV algebra flags to `true`. The flag `disputeSettledByCurrentStage` remains `false`.

The current code base is large enough to distinguish three meanings of "available". First, pure ordered-real consequences are proved outright. Second, route theorems prove that certain named source-facing records imply the desired endpoint. Third, the records themselves remain obligations when they stand for Hodge-theater, Frobenioid, log-Kummer, hull, determinant, or multiradial constructions not yet formalized from IUT I–III. Only the first two meanings are currently achieved.

8 What Is Not Yet Proved

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The following mathematics is still external to the code:

1. initial Θ-data and its arithmetic hypotheses from IUT I;

2. Hodge theaters, Frobenioids, log-shells, prime strips, and theta links;

3. the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;

4. the construction of the actual (Ind1), (Ind2), (Ind3) data in Theorem 3.11;

5. the genuine holomorphic hull and determinant constructions used in Step (xi), beyond the present ordered-real upper-ray model;

6. the proof that the typed SHE and Hodge-descent transport obligations follow from Mochizuki's source constructions;

7. the construction of the input prime-strip link (IPL), simultaneous holomorphic expressibility (SHE), and algorithmic parallel transport (APT) properties as actual consequences of Theorem 3.11 rather than records;

8. the arithmetic-divisor, height, log-different, and log-conductor constructions behind the IUT IV Diophantine applications.

9 Implementation Audit

The code contains no `sorry`, `admit`, or Lean axioms in the Stage 1 corridor. The current risk is not hidden unsoundness of this form, but mathematical incompleteness: several records are deliberately named obligation interfaces. These interfaces are acceptable only as temporary boundaries. The next phase must replace them by constructions from the source papers or mark them as unresolved.

The largest engineering issue is `Iut/Stage1/IUTStage1Source.lean`, which currently contains most of the finite-label, Gaussian, Hodge/SHE, hull, Step (x), Step (xi), Step (xii), and IUT IV algebra. This is now too large for comfortable review. Low-risk future refactors are:

finite labels and q^{j^2} averages	separate module,
Step (xi) hull/determinant upper rays	separate module,
Step (xii) local Frobenioid shifts	separate module,
IUT IV real algebra	separate module.

These refactors should be mechanical only: move declarations, preserve names where possible, and rebuild after each move.

10 Next Work

The next mathematical milestone is to construct, rather than assume, more of the source-facing SHE and Hodge-descent preservation data. The immediate route is:

$$\begin{gathered}
q^{j^2} \text{ finite Gaussian endpoint} \\
\Downarrow \\
\text{Gaussian and splitting monoid construction} \\
\Downarrow \\
\text{factored square/full-label preservation} \\
\Downarrow \\
\text{Hodge/SHE transport audit} \\
\Downarrow \\
\text{hull/log-volume comparison} \\
\Downarrow \\
C_\Theta \geq -1.
\end{gathered}$$

The project should continue by replacing the present record fields for factored SHE preservation and Hodge-descent source-target alignment with the corresponding constructions from IUT I–III.

The practical priority order is:

1. factor the large source module into reviewable finite-label, hull, Frobenioid-shift, and IUT IV algebra modules without changing theorem statements;

2. extend the Gaussian-derived factored SHE construction beyond the finite degree-evaluation and canonical-label shadows toward actual Gaussian/Frobenioid material from IUT II;

3. derive the remaining real equalities in the 1-column log-Kummer upper-semi compatibility object from the actual log-Kummer construction of IUT III, Step (x);

4. replace the hull/determinant endpoint records by the actual holomorphic-hull and determinant operations from Remark 3.9.5 and Step (xi);

5. only then evaluate whether the Stage 1 interface has enough source content to bear on the Mochizuki–Scholze–Stix disagreement.

References

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[3] S. Mochizuki, *Inter-universal Teichmuller Theory III: Canonical Splittings of the Log-theta-lattice.*

[4] S. Mochizuki, *Inter-universal Teichmuller Theory IV: Log-volume Computations and Set-theoretic Foundations.*

[5] S. Mochizuki, *On the Formalization of IUT: Preliminary Progress Report*, 2026.

[6] P. Scholze and J. Stix, *Why abc is still a conjecture.*